

# Global Retail Profitability & Business Performance Analytics

## Project Objective

The objective of this project is to develop an end-to-end Business Intelligence solution using SQL Server and Power BI to analyze retail operations and provide actionable insights that improve profitability, optimize business performance, and support data-driven decision-making.

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## Core Business Problem

Although the company generates significant sales across multiple countries and product categories, management wants to understand whether these sales are translating into sustainable profits. The business seeks to identify opportunities to improve profitability by analyzing customers, products, discounts, shipping costs, regions, and sales performance.

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## Key Business Questions

### 1. Executive Performance

**Business Question:**

How is the business performing overall?

**Analysis Focus:**

- Total Sales
  - Total Profit
  - Profit Margin
  - Total Orders
  - Total Customers
  - Average Order Value
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### 2. Profitability Optimization

**Business Question:**

Why is profit not increasing at the same rate as sales?

**Analysis Focus:**

- Profit trends
  - Loss-making transactions
  - Discount impact
  - Shipping cost impact
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### 3. Product Portfolio Analysis

**Business Question:**

Which products and categories contribute the most to profit, and which products generate losses?

**Analysis Focus:**

- Top-performing products
  - Loss-making products
  - Category performance
  - Sub-category performance
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### 4. Customer Value Analysis

**Business Question:**

Who are the most valuable customers and customer segments?

**Analysis Focus:**

- High-value customers
  - Customer profitability
  - Segment analysis
  - Revenue contribution
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### 5. Geographic Performance Analysis

**Business Question:**

Which countries, markets, and regions contribute most to business success?

**Analysis Focus:**

- Country performance
- Regional performance
- Market performance

- Geographic profitability
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## 6. Discount Strategy Evaluation

### **Business Question:**

Are discounts driving sales while reducing profitability?

### **Analysis Focus:**

- Discount vs Sales
  - Discount vs Profit
  - Discount effectiveness
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## 7. Shipping Cost Efficiency

### **Business Question:**

How do shipping costs and shipping methods affect profitability?

### **Analysis Focus:**

- Shipping Cost Analysis
  - Ship Mode Analysis
  - Operational efficiency
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## 8. Time-Based Performance Analysis

### **Business Question:**

How has business performance changed over time?

### **Analysis Focus:**

- Yearly Sales
  - Yearly Profit
  - Yearly Orders
  - Weekly Sales
  - Weekly Profit
  - Weekly Order Trends
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## 9. Operational Risk Identification

**Business Question:**

Which business areas consistently underperform and require management attention?

**Analysis Focus:**

- Loss-making products
  - Underperforming categories
  - Weak-performing regions
  - Low-profit segments
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## 10. Strategic Business Recommendations

**Business Question:**

What actions should management take to improve profitability and long-term business performance?

**Expected Recommendations:**

- Optimize discount policies
  - Promote high-margin products
  - Improve regional strategy
  - Reduce operational costs
  - Focus on high-value customers
  - Improve overall profitability
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## Technical Scope

### SQL Server

- Data Cleaning
- Data Validation
- Dataset Profiling
- KPI Calculations
- Exploratory Data Analysis (EDA)
- Business Analysis
- Analytical Views

### Power BI

- Executive Dashboard
- Profitability Dashboard
- Product Performance Dashboard
- Customer Analysis Dashboard

- Regional Performance Dashboard
  - Shipping & Operations Dashboard
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## **Skills Demonstrated**

### **Technical Skills**

- SQL Server
- Data Cleaning & Validation
- Aggregate Functions
- Window Functions
- Common Table Expressions (CTEs)
- Views
- Power BI
- DAX
- Dashboard Development

### **Business Skills**

- Executive KPI Reporting
  - Profitability Analysis
  - Customer Analytics
  - Product Analytics
  - Geographic Analysis
  - Operational Analysis
  - Business Intelligence
  - Data-Driven Decision Making
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## **Expected Business Impact**

This project enables management to:

- Improve profitability
- Identify high-performing products and customers
- Optimize discount strategies
- Reduce operational inefficiencies
- Improve regional decision-making
- Support strategic business growth through data-driven insights